[CSED233] Data Structure

AVL Tree

Seung-Hwan Baek

POSTECH

Balanced Trees

- Tree height is $O(\log n)$
  - No leaf is much further away from the root than any other leaf
  - To minimize the longest path from the root to any leaf node

- Why balanced?
  - Insert, Delete, and Member take $O(\log n)$ time

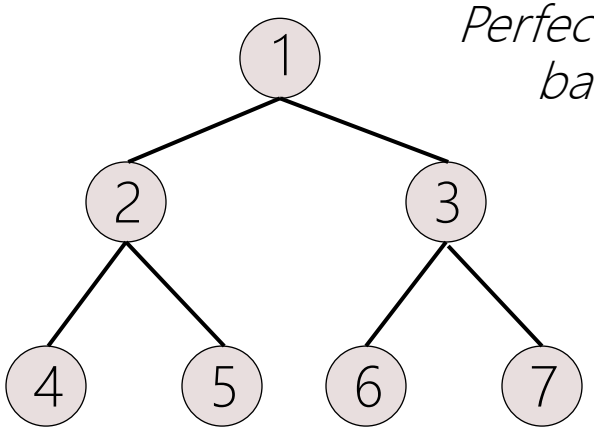
Data Structure	Worst case	Average
Binary Search Tree (BST)	$O(n)$	$O(\log n)$
Balanced Search Trees	$O(\log n)$	$O(\log n)$

Height-Balanced

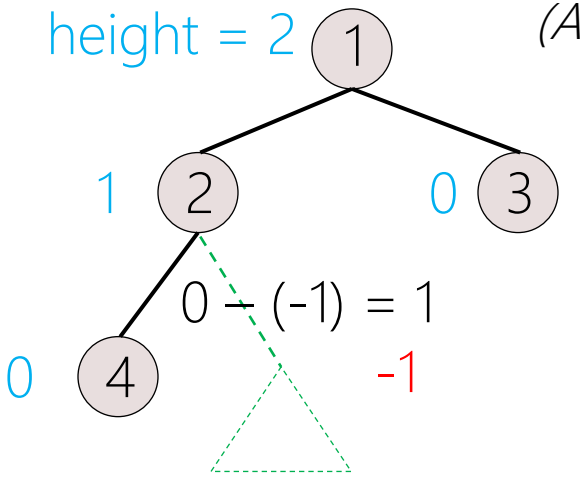
- Perfectly height-balanced
  - If the sub-trees of any node are of the same height, so that all leaf nodes are at the same level
  - Too tough condition to satisfy

- ('Almost' perfectly) height-balanced
  - For each node, the height of its sub-trees can differ by at most 1, & the sub-trees are also height-balanced
  - We can prove that height = $O(\log n)$ since it is an almost balanced tree

Example: Height-Balanced

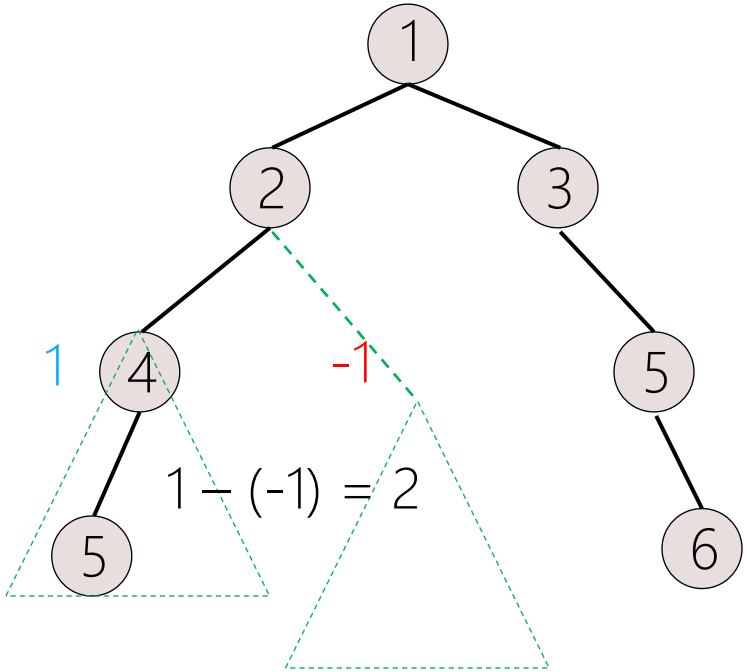


Perfectly height-balanced



(Almost) height-balanced

Not height-balanced



*Height for non-existing subtree is assumed to be -1

AVL Tree

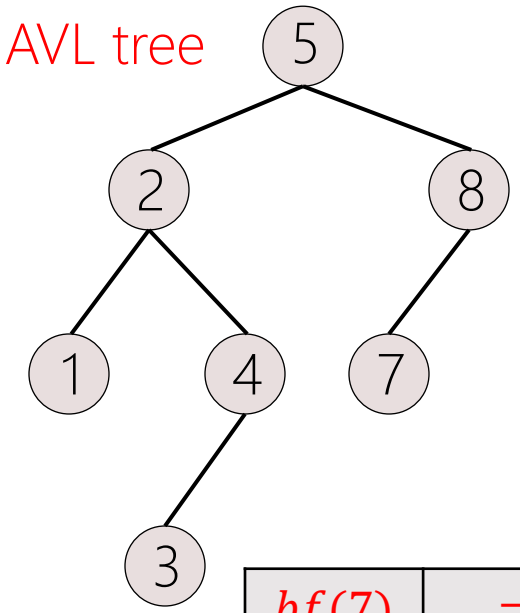
- Height-balanced BST
  - For every node in the tree, the left & right sub-trees differ in height by at most 1

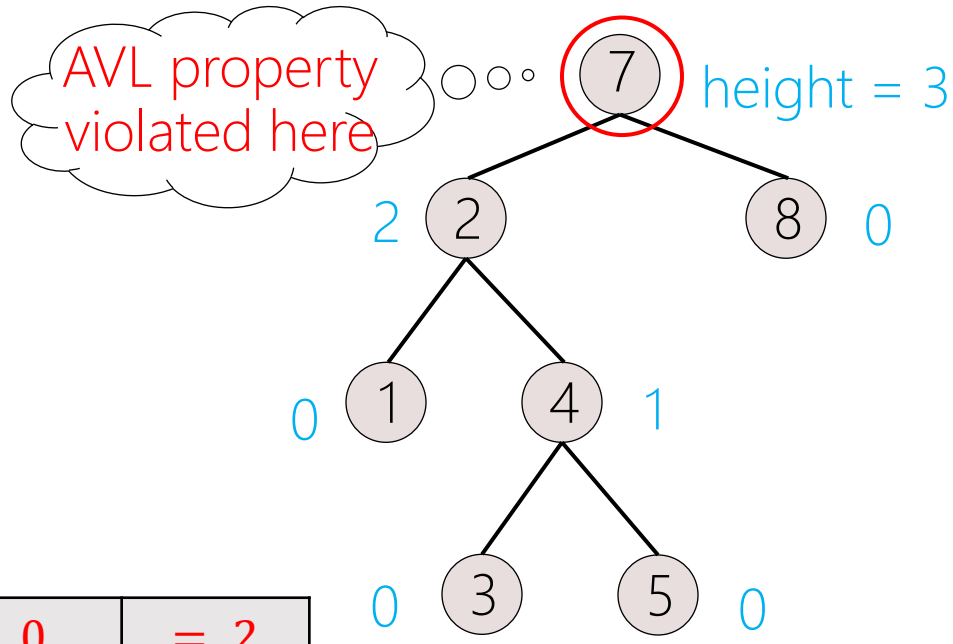
- Balance factor of node x
  - $bf(x) = height(left(x)) - height(right(x))$
  - Note:
    - $height(one\ node) = 0, height(empty\ tree) = -1$

- AVL tree property
  - A valid AVL tree must satisfy: $-1 \leq bf(x) \leq 1$ for every node x

AVL Tree & Balance Factors

- An invalid AVL tree can be detected by $bf(x)$





$bf(7)$	$= 2 - 0$	$= 2$
$bf(2)$	$= 0 - 1$	$= -1$
…	……	…
$bf(3)$	$= (-1) - (-1)$	$= 0$
$bf(5)$	$= (-1) - (-1)$	$= 0$

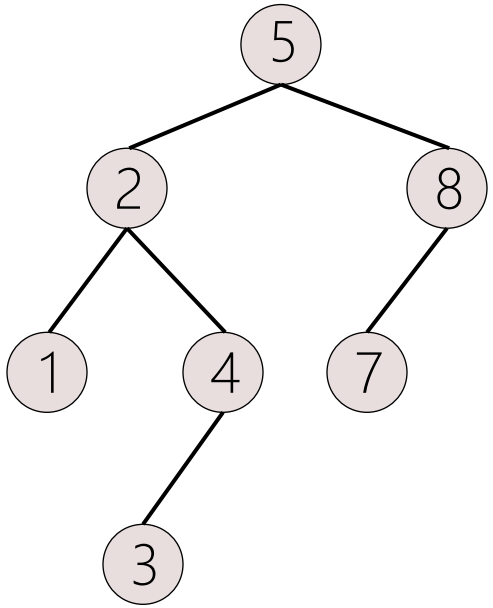
Quiz

- Suppose a binary tree has height $O(\log n)$.
- Does this mean the tree is balanced in the AVL sense?

Insertion in AVL tree

- Insertion can be done as in BST
  - But, it may cause violation of AVL tree property (height balanced)

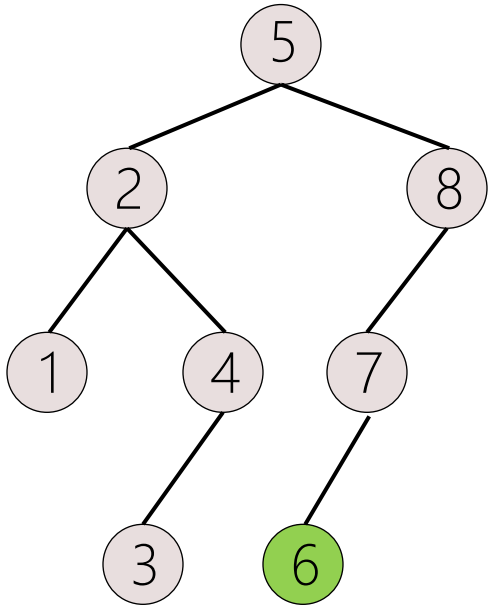
- Restore the AVL property if needed
  - $Insert(6)$



Insertion in AVL tree

- Insertion can be done as in BST
  - But, it may cause violation of AVL tree property (height balanced)

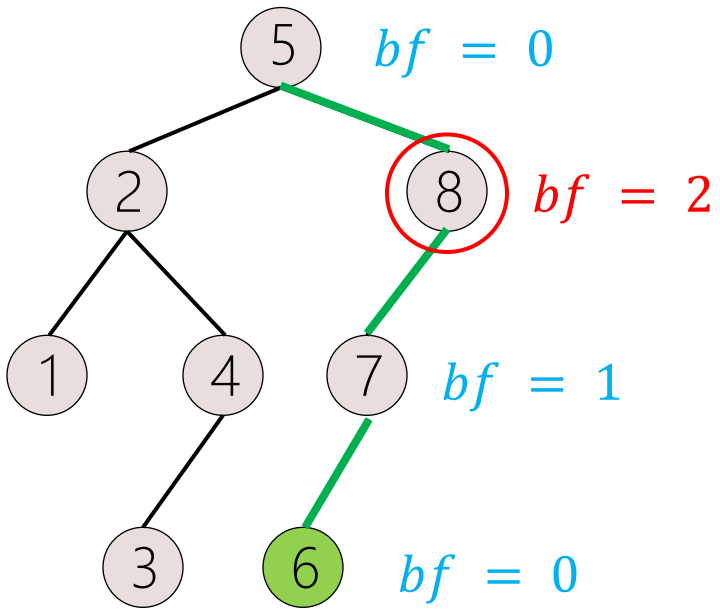
- Restore the AVL property if needed
  - After $Insert(6)$



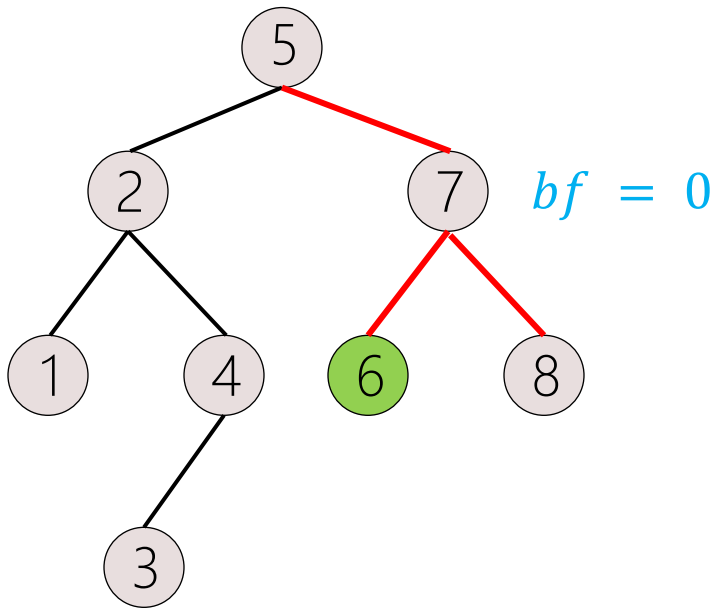
Insertion in AVL tree

- Insertion can be done as in BST
  - But, it may cause violation of AVL tree property (height balanced)

- Restore the AVL property if needed
  - After $Insert(6)$, violated at 8

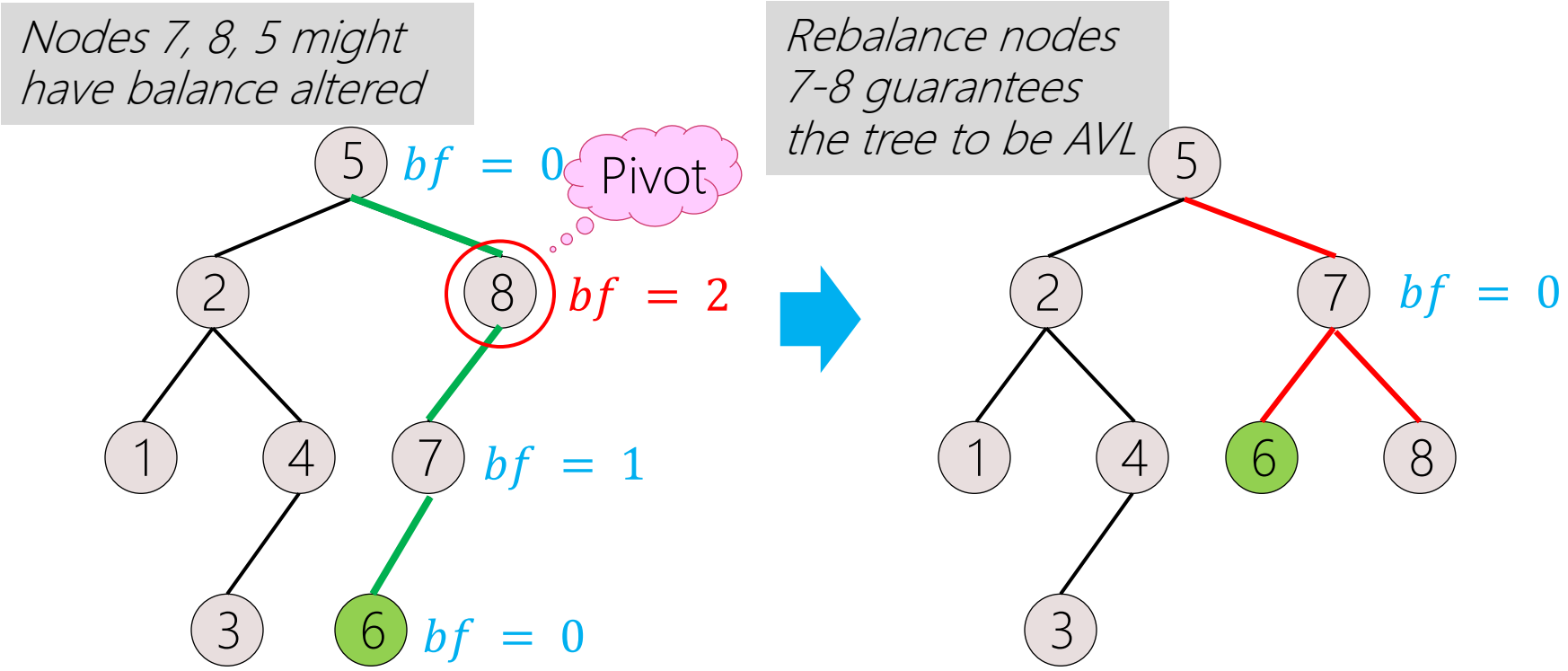


- Restore AVL property



Some Observations

- After insertion,
  - Only nodes along the path from insertion point to root might have their balance altered
  - Need to rebalance the tree at such the "deepest" node (called *pivot*)



Pivot Node & Rotation

- Pivot node (denoted by A)
  - The nearest ancestor of the newly inserted node that may go out of balance
    - The "deepest" unbalanced node from the root
  - $bf(A)$ becomes $+2$ or -2 after insertion

- Restoring the balance of the pivot node
  - Restores the balance of the whole sub-tree & potentially all of the nodes that were affected by the insertion
  - This restructuring mechanism is called "rotation"

Imbalance Types

- Let A be the pivot node
  - The deepest node that violates the AVL property

- Only four possible types:
  - LL: insertion into left sub-tree of left child of A
    - The insertion took place in the *left sub-tree* of a node whose $parent(A)$ was *left high*
  - RR: symmetric case of LL
  - LR: insertion into right sub-tree of left child of A
    - The insertion took place in the right sub-tree of a node whose $parent(A)$ was *left high*
  - RL: symmetric case of LR

Tree Rotations: Two Types

- Single rotation (for LL or RR imbalance)
  - Requires only a single rotation on the pivot node as an axis

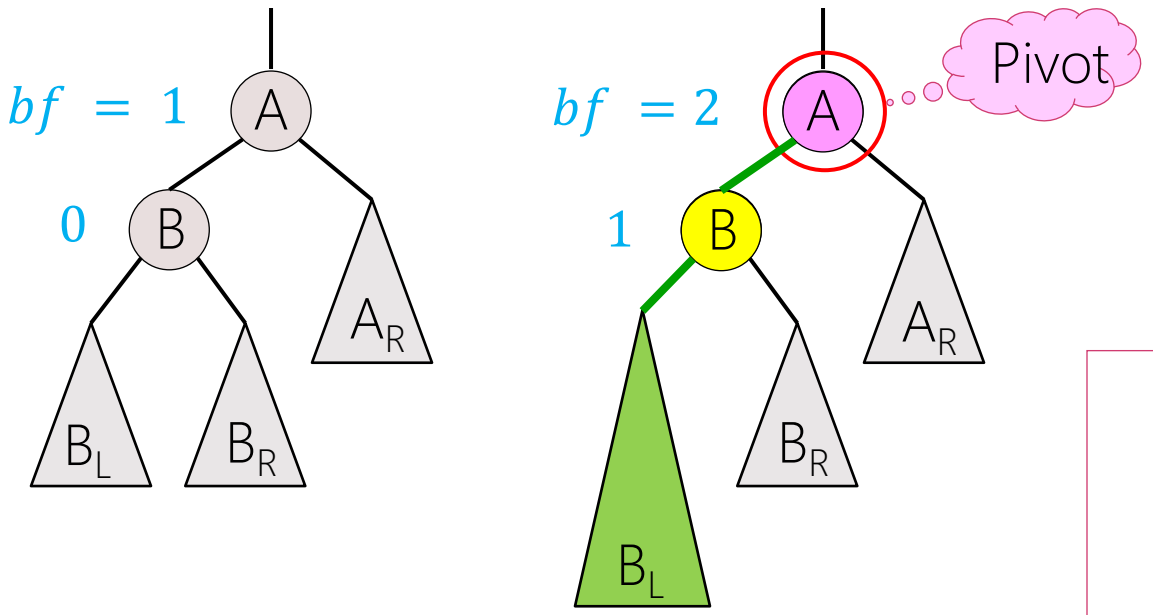
- Double rotation (for LR or RL imbalance)
  - Requires a prior rotation on the root of the affected sub-tree, then a rotation on the pivot node
    - LR = RR + LL
    - RL = LL + RR
  - The first rotation does not affect the degree of imbalance but convert it to a simple LL or RR

Insertion Algorithm

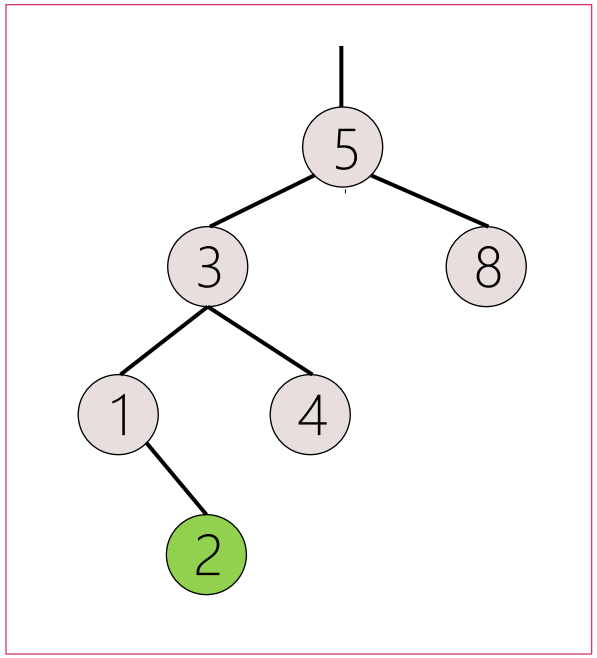
- First, insert a new key as in BST
- Then, trace the path from the new leaf towards the root:
  - For each node x, check if $-1 \leq bf(x) \leq 1$
    - If yes, proceed to $parent(x)$
    - If no, rotate the tree on node x, and then finish (early termination)
      - That's the pivot

- Note:
  - Once we perform a rotation on node x, we don't need to do any rotation on any ancestor of x
  - Rotation may change x, so connect the resulting tree to $parent(x)$
  - Update the height of nodes involved in rotations

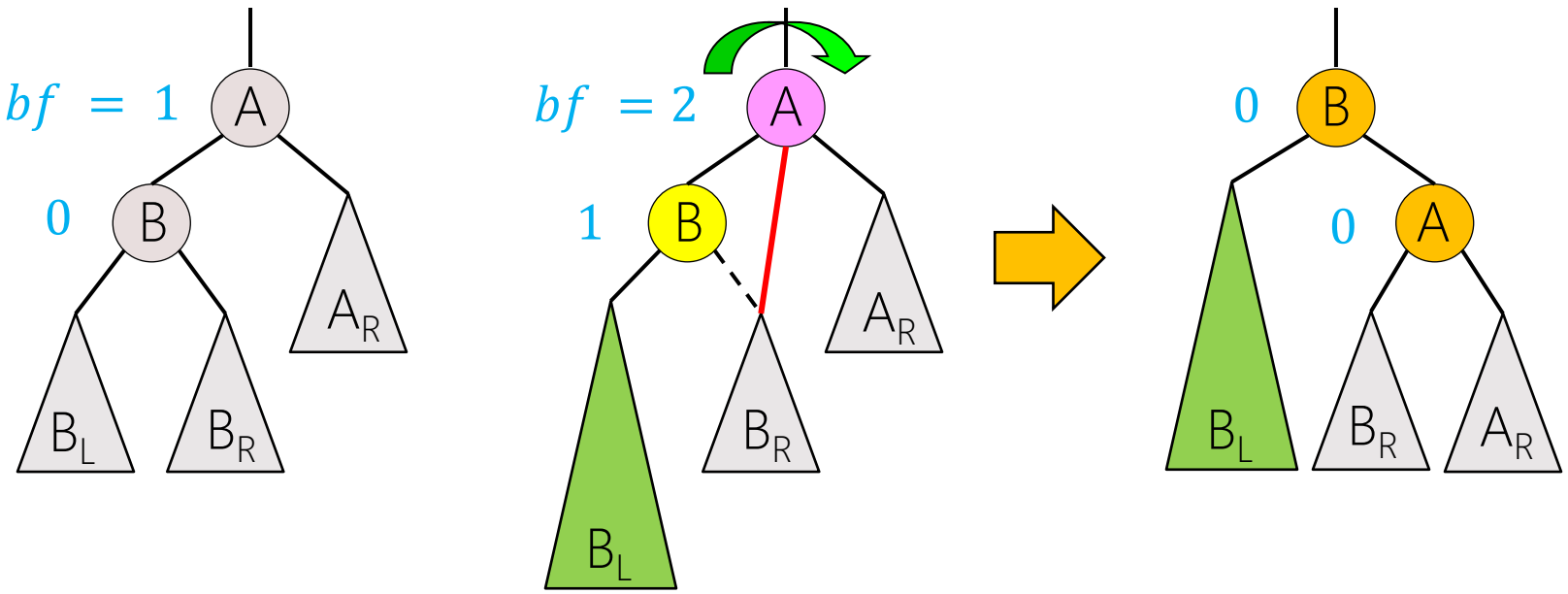
Single Rotation: LL Case



Before insertion *After insertion*



Single Rotation: LL Case (Right Rotation)

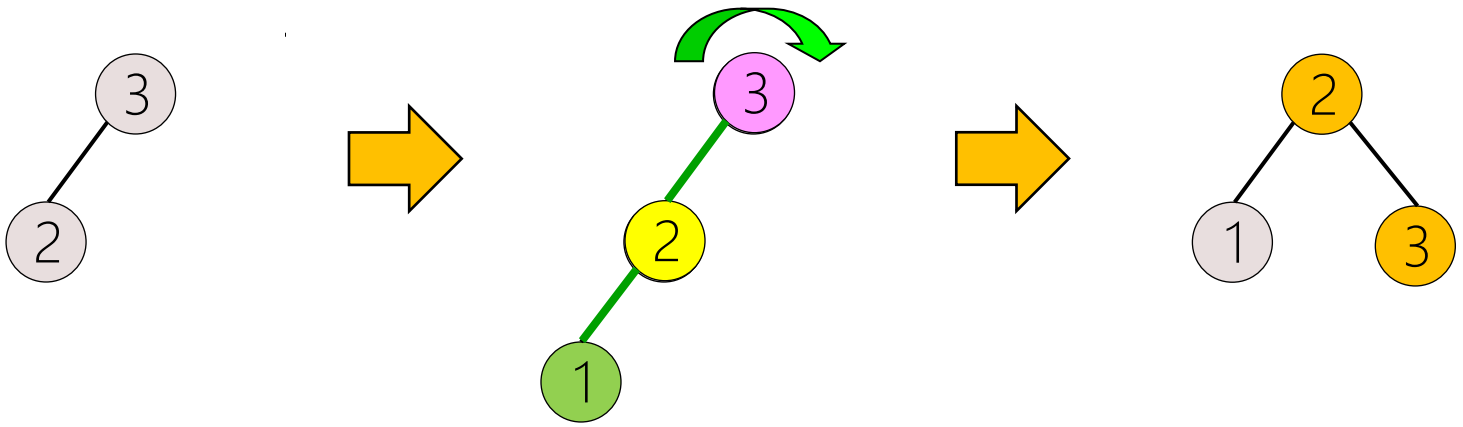


Before insertion *After insertion* *After LL (Right) rotation*

- A single rotation takes $O(1)$ time
  - Insertion takes $O(Height\ of\ AVL\ tree)$ time
- RR case is symmetric to LL case

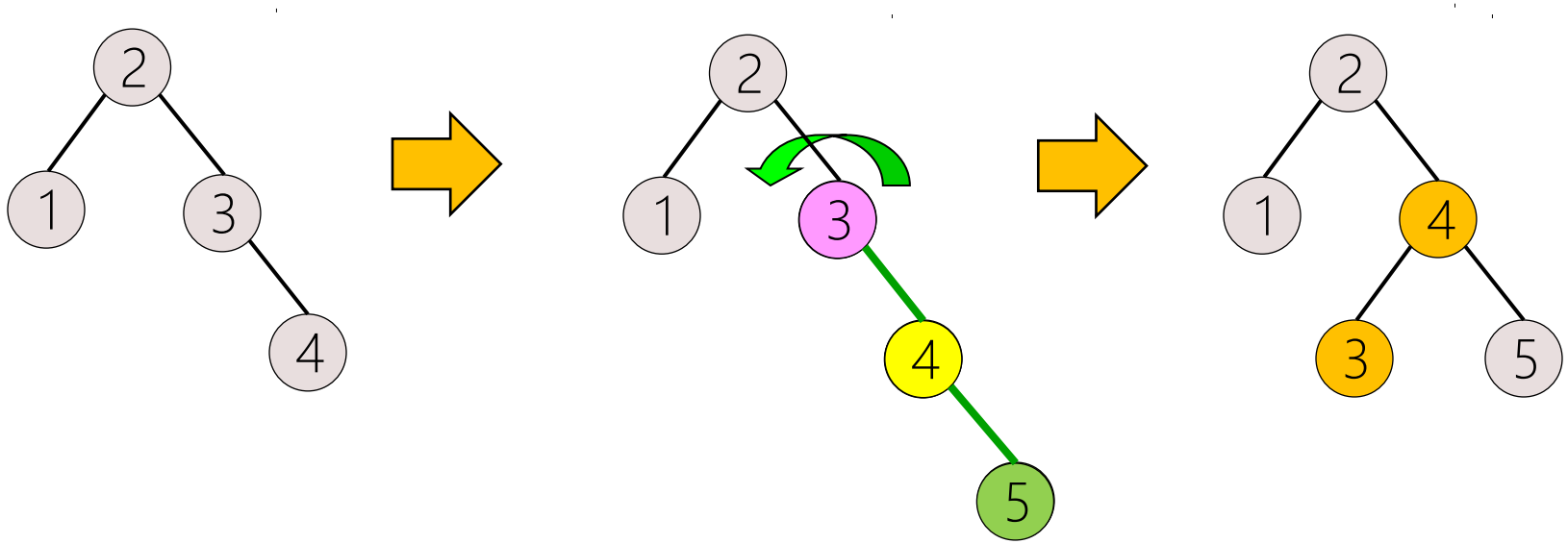
Single Rotation: Example

- Insert 3, 2
- Insert 1
  - Violation at node 3
  - Right (LL) rotation



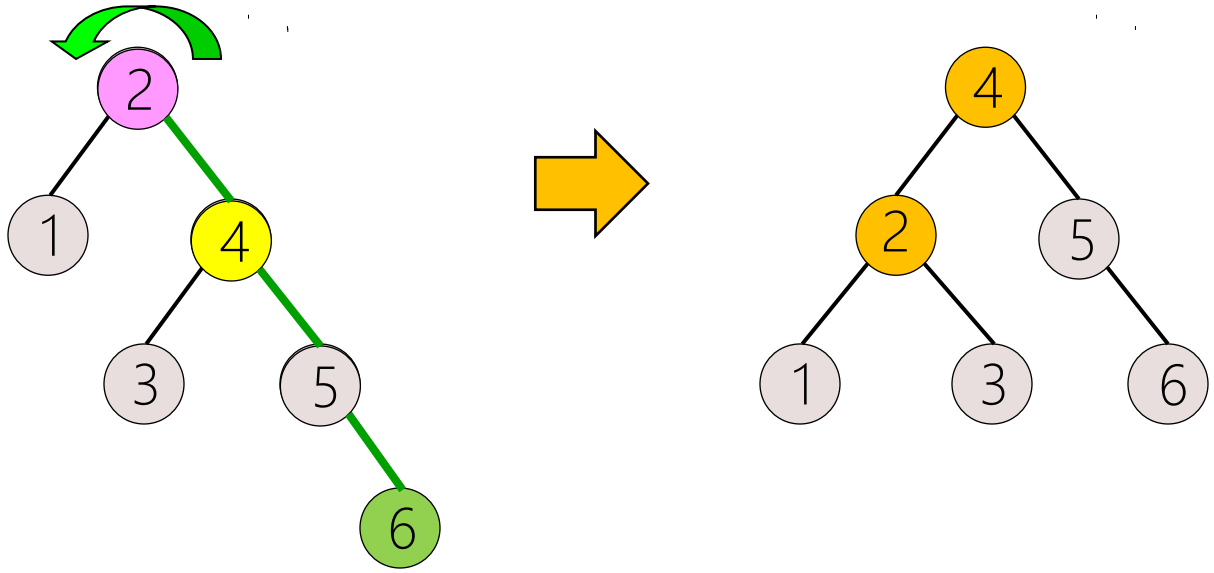
Single Rotation: Example

- Insert 4
- Insert 5
  - Violation at node 3
  - Left (RR) rotation



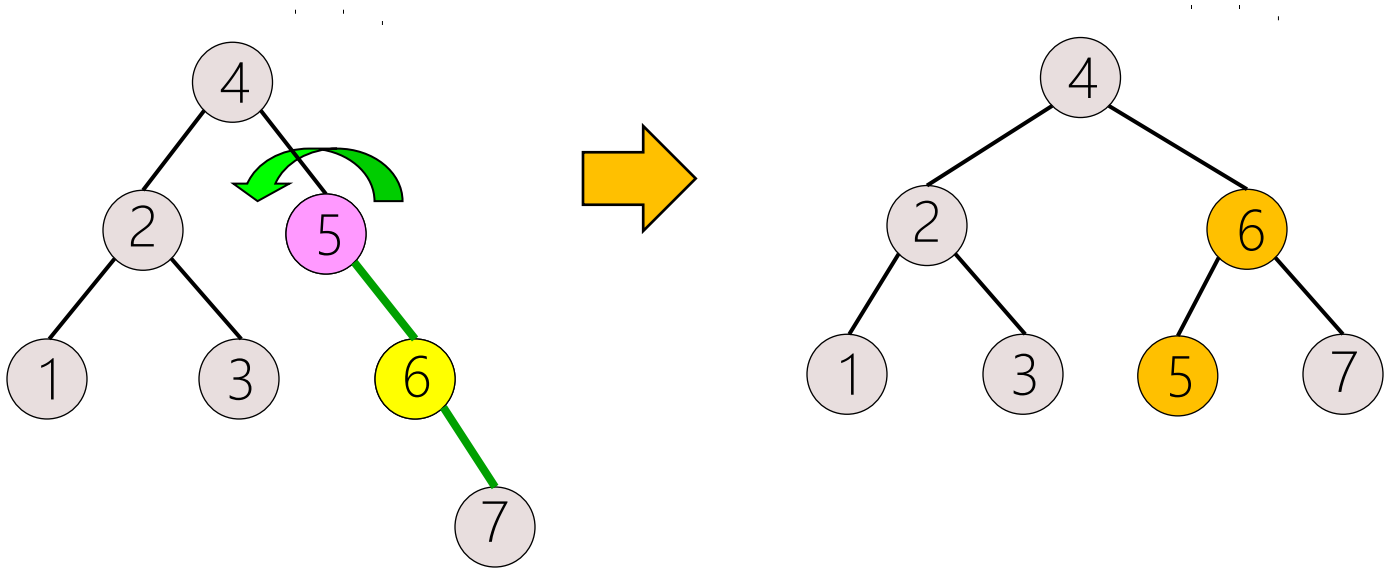
Single Rotation: Example

- Insert 6
  - Violation at node 2
  - Left (RR) rotation



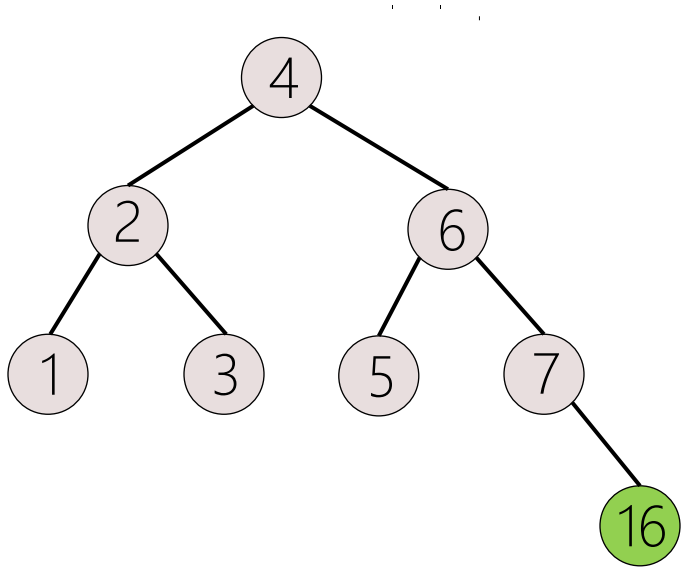
Single Rotation: Example

- Insert 7
  - Violation at node 5
  - Left (RR) rotation

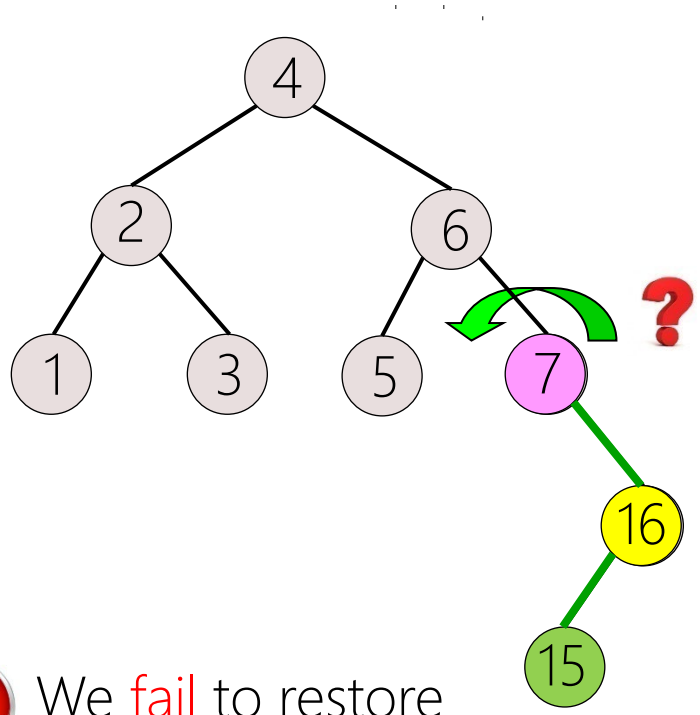


Single Rotation: Example

- Insert 16

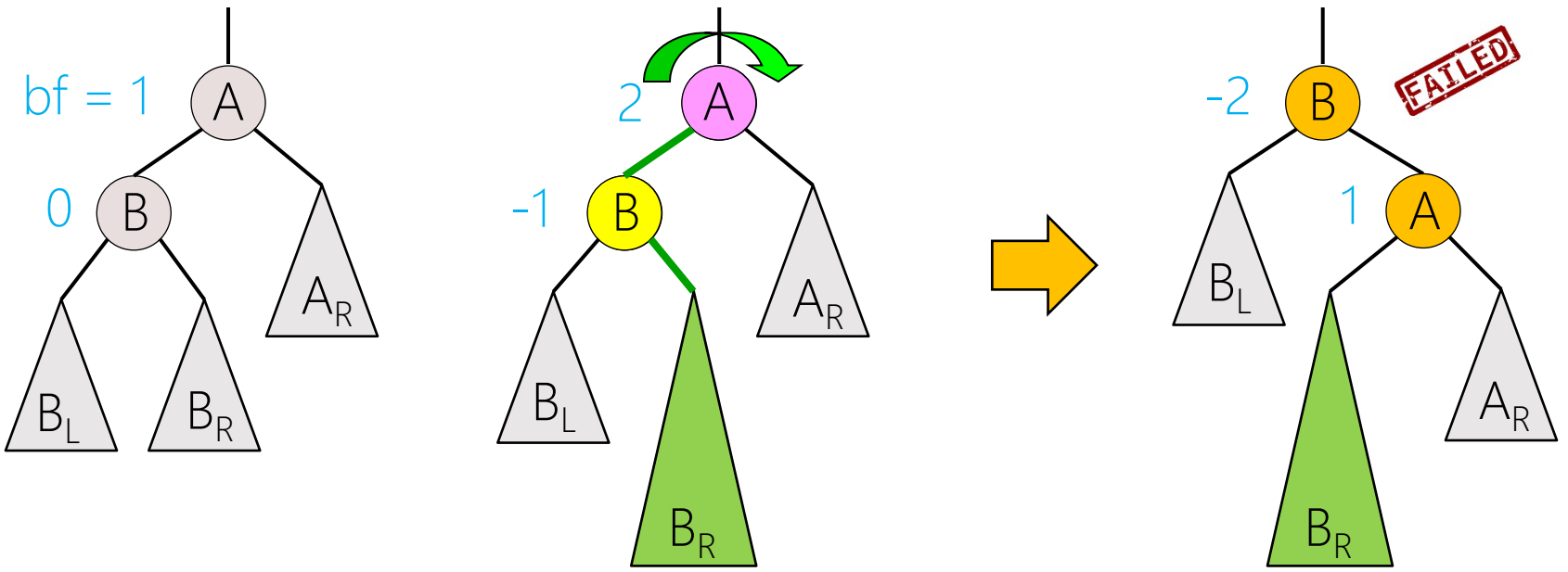


- Insert 15
  - Violation at node 7



We fail to restore the balance by single rotation

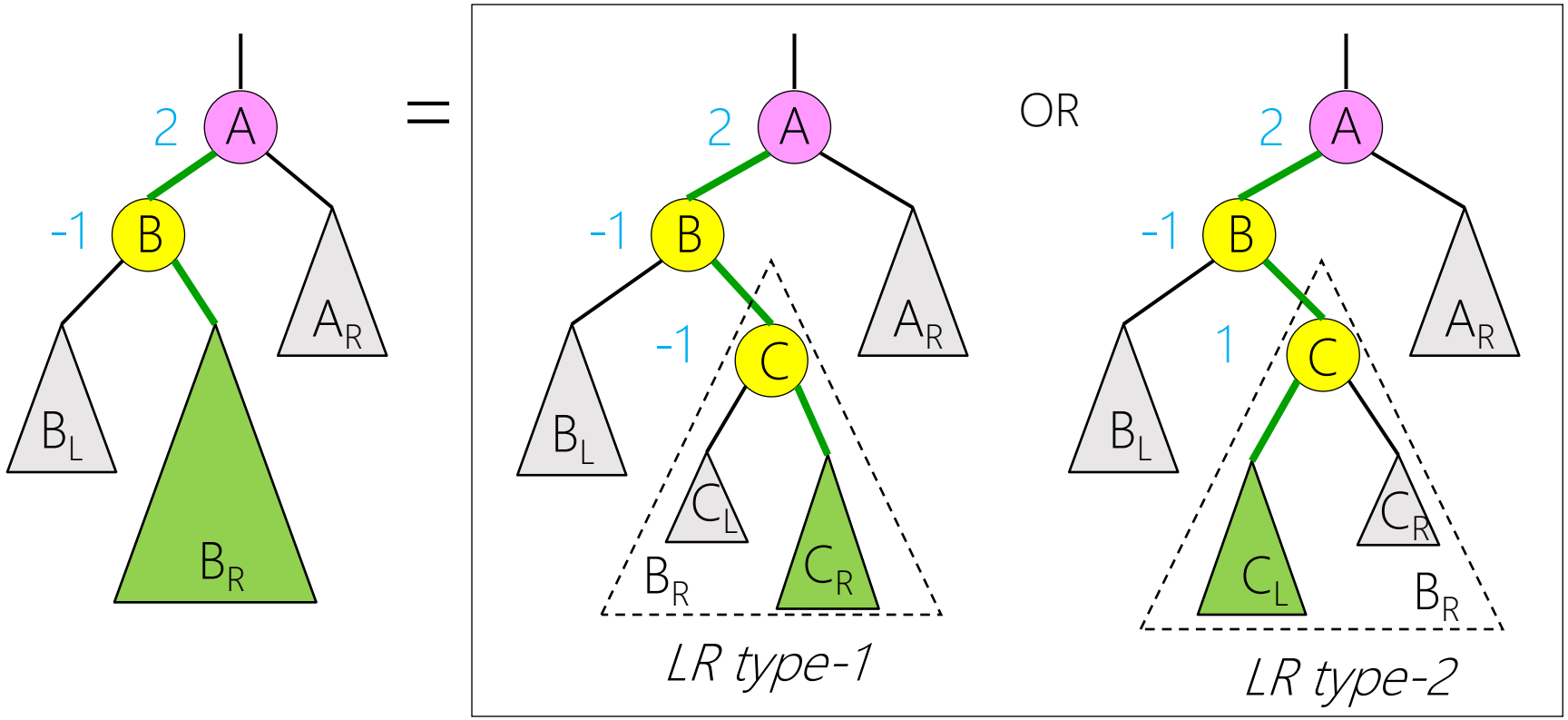
Double Rotation: LR case



Before insertion *After insertion (LR type)* *After right rotation*

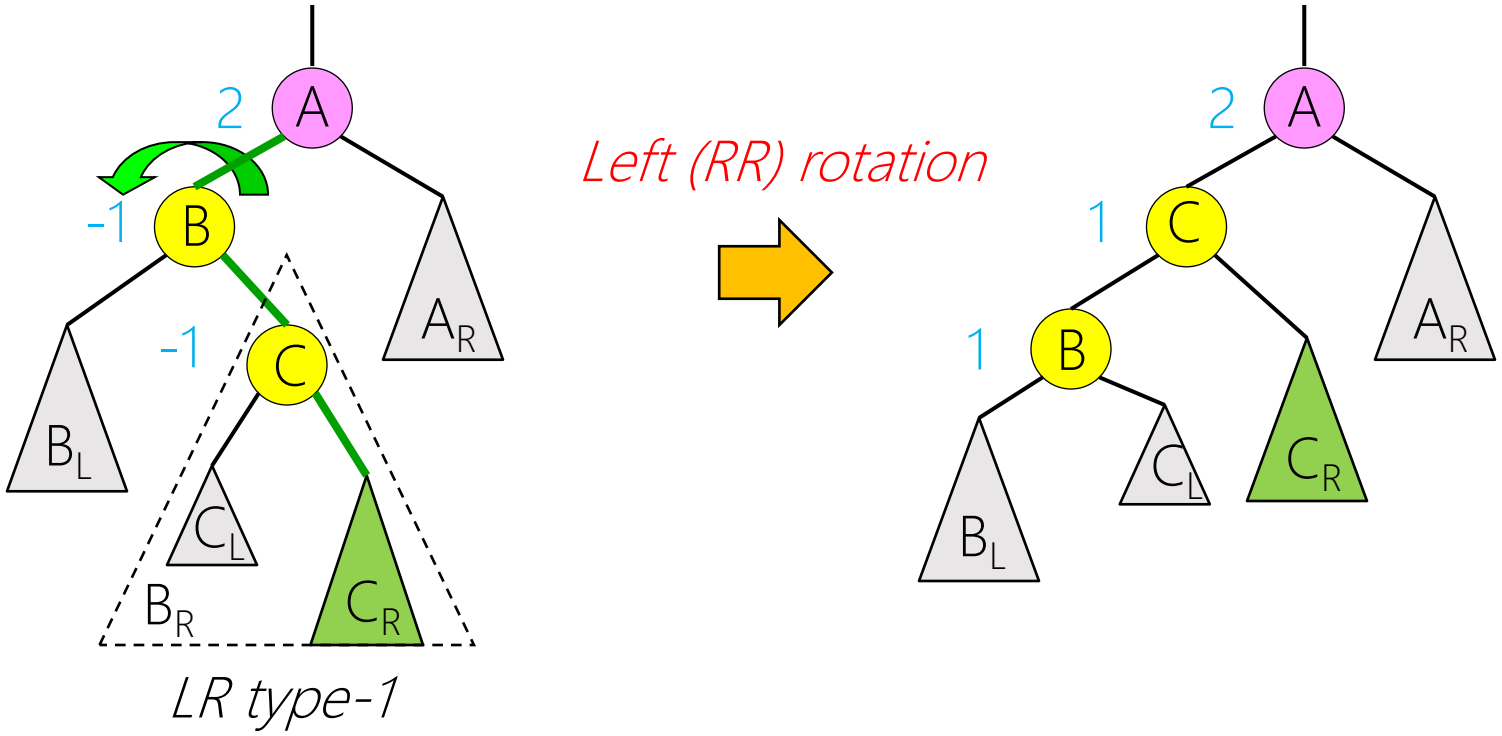
Single rotation only fails to restore the balance

Double Rotation: LR case-1 & case-2



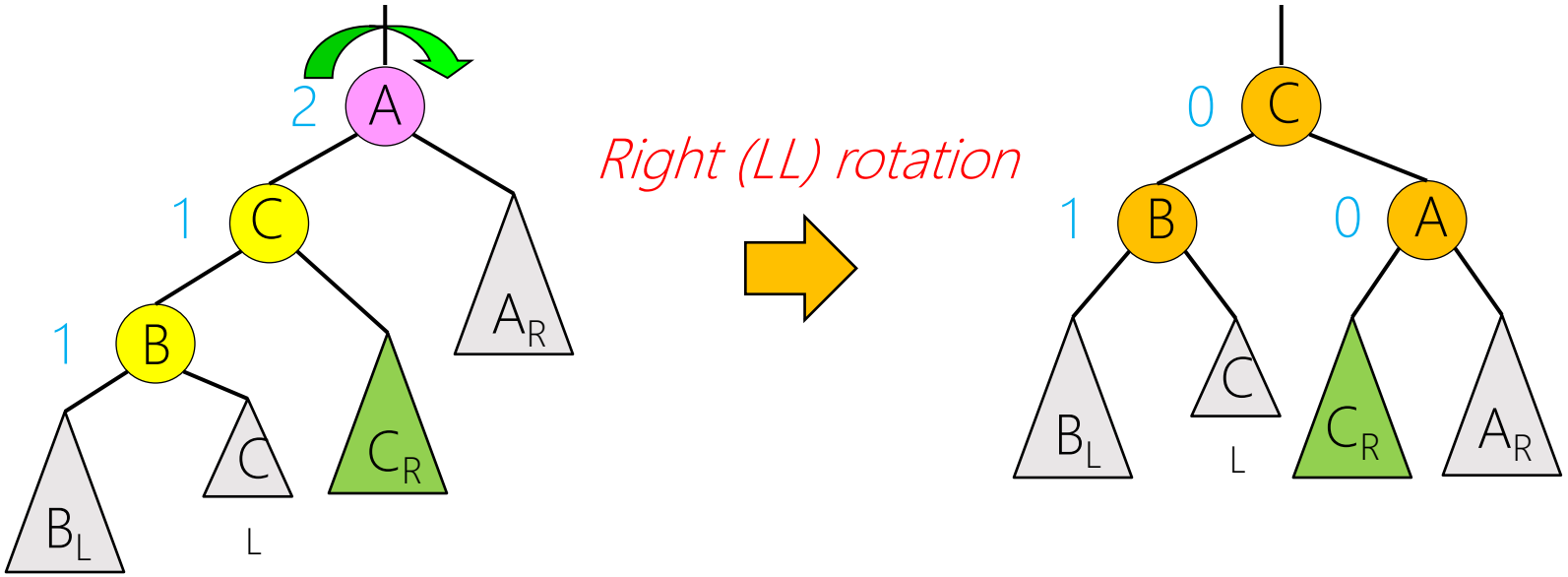
- Further examine the sub-tree B_R
  - A new node is inserted either in C_L or C_R

Double Rotation (LR): First Rotation (RR)



- First, do a single left rotation on node B as an axis (as in RR case)

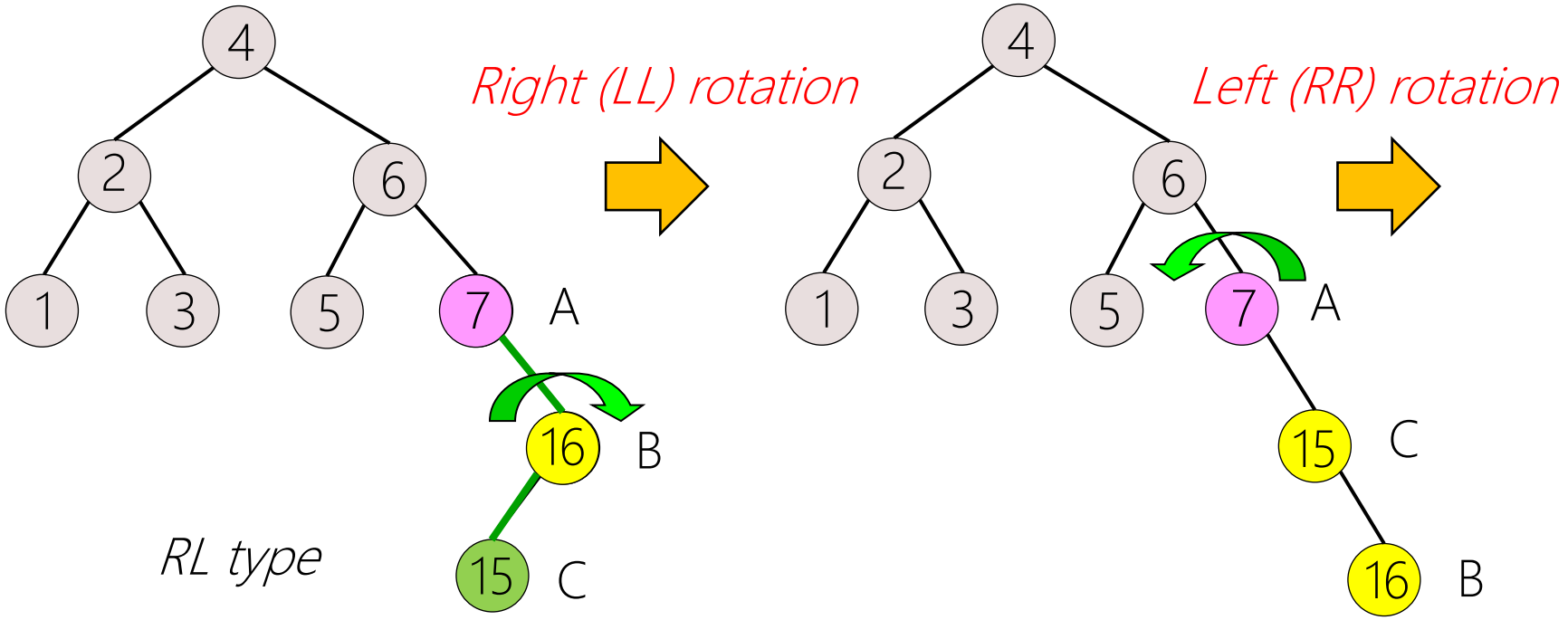
Double Rotation (LR): Second Rotation (LL)



- Then, do a single right rotation at node *A* (as in LL case)
  - Consequently LR = RR + LL : double rotation
- RL case is symmetric to LR case (RL = LL + RR)

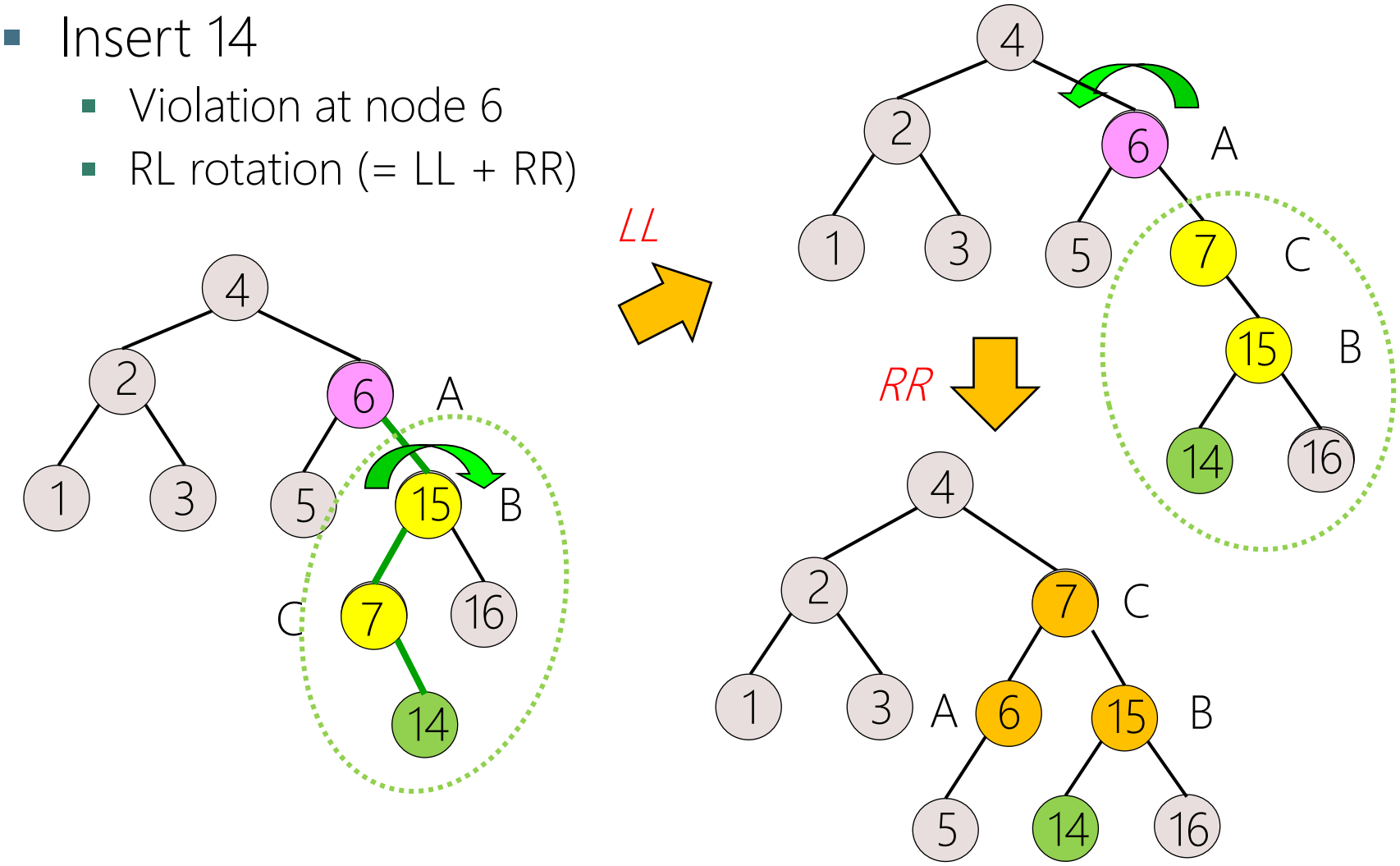
Double Rotation: Example

- Restart our example
  - We've inserted 3, 2, 1, 4, 5, 6, 7, 16, but we failed to insert 15
  - We'll insert 15 again



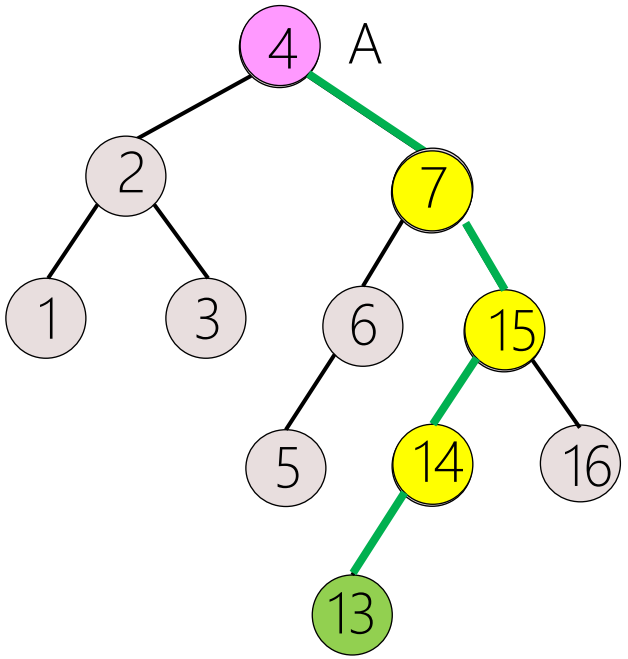
Single Rotation: Example

- Insert 14
  - Violation at node 6
  - RL rotation (= LL + RR)



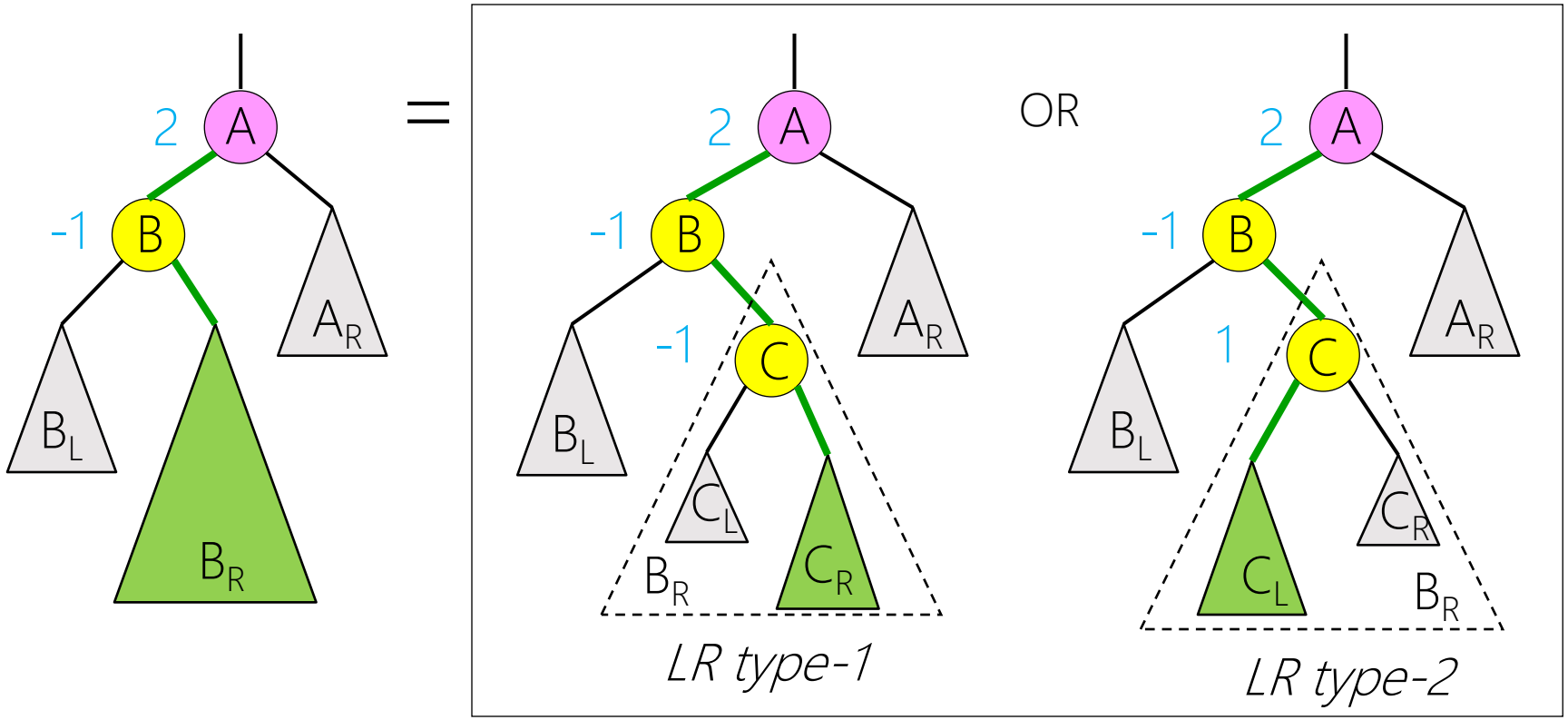
Single Rotation: Example

- Insert 13, 12, 11, 10, 8, 9
  - Violation at node 4
- What's the final results?

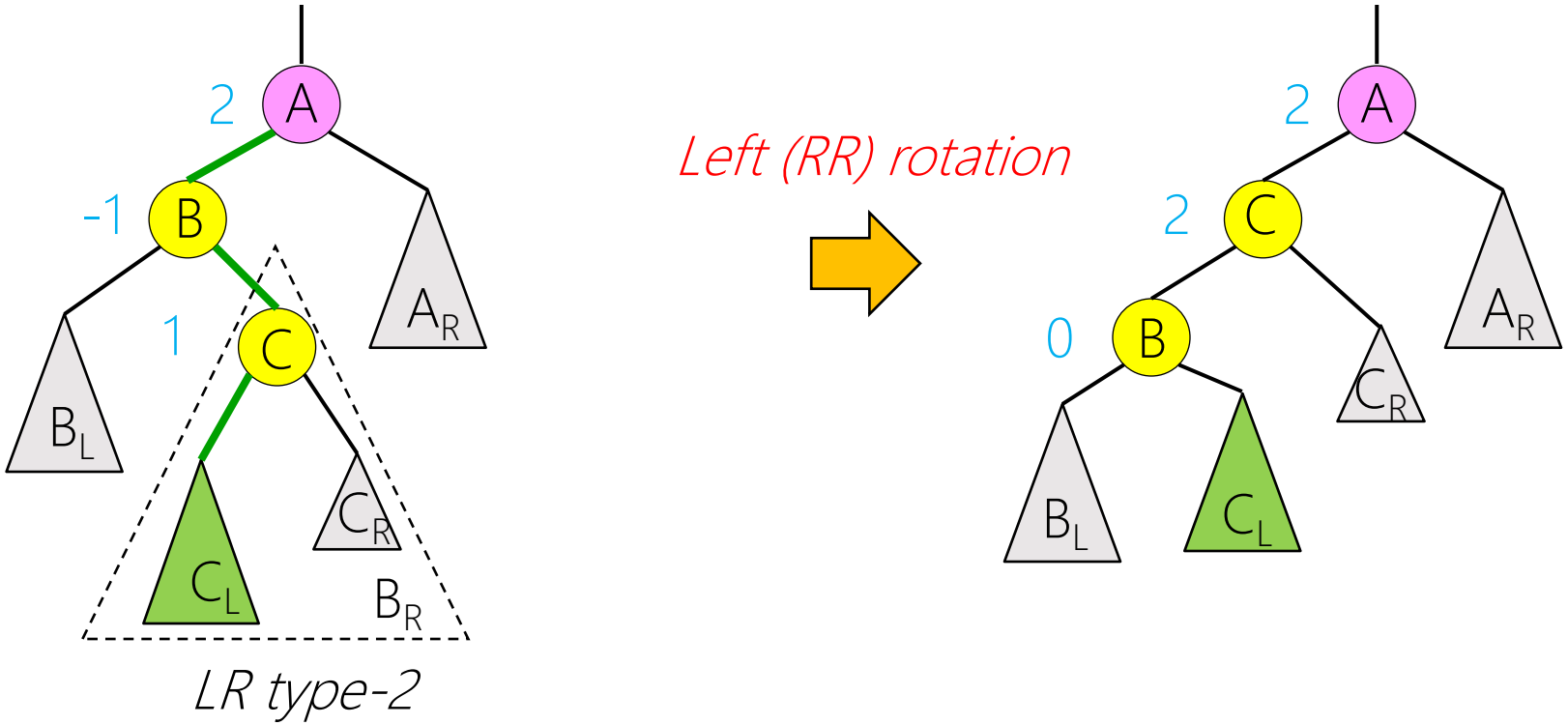


.....

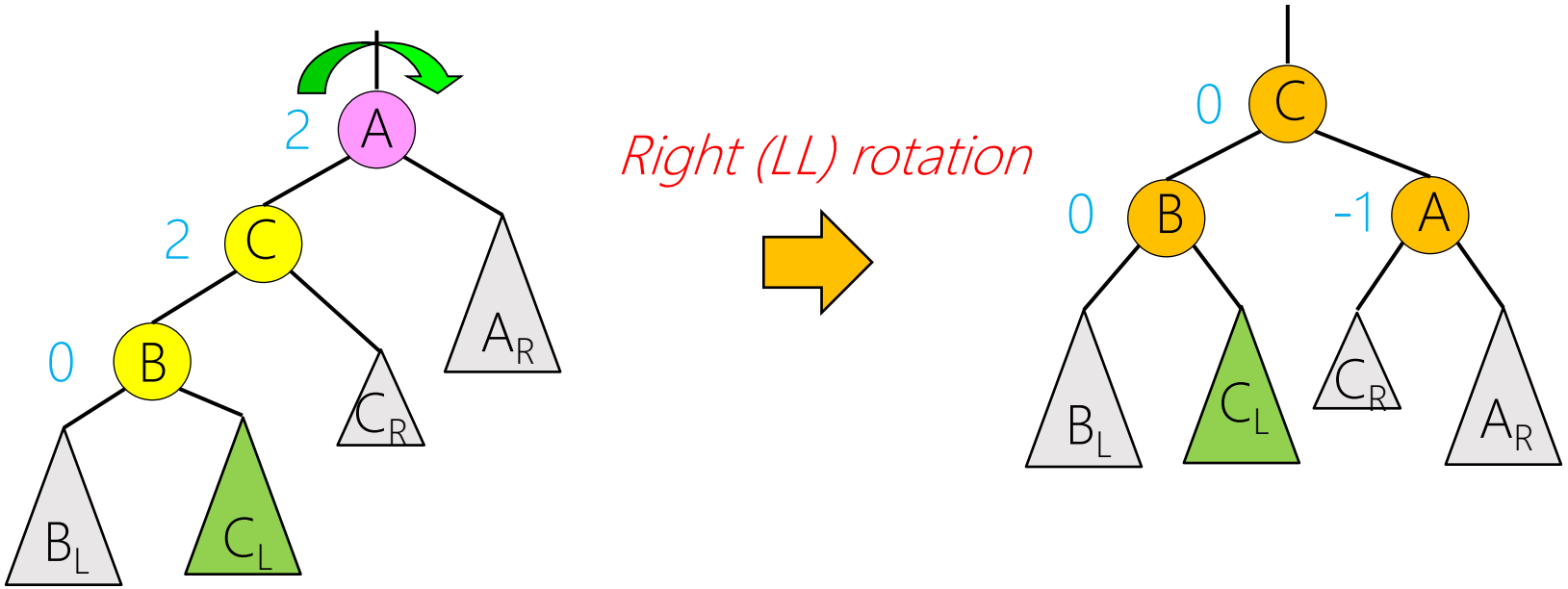
What about case-2?



What about case-2?



What about case-2?



Quiz

- Insert 3, 2, 1, 4, 5, 6, 7 into an AVL tree
- How many rotations happened?

Quiz

- During AVL insertion, once a rotation is performed at the pivot node, why is it correct to stop and never rebalance any ancestor?

Time Complexity: AVL Insertion

- Time complexity of each step of AVL insertion
  - BST insertion
    - $O(\log n)$
  - Checking & rotation for all ancestors of the inserted node
    - $O(\log n)$
    - Checking & rotation (if needed) for a single node
      - To update the balance factor of a single node = $O(1)$
      - Rotation = $O(1)$
- Overall time complexity of insertion is: $O(\log n)$

Review: Deletion in BST

- There are 3 cases of deletion from
  - *Leaf* node
  - *Degree-1* node (with one child)
  - *Degree-2* node (with two children)
    - The smallest key must be in a *leaf* or *degree-1* node

- In all cases, the last (deepest) deleted node is a
  - *leaf* node or *degree-1* node

Deletion from AVL Tree

- First, perform BST deletion
- Then, trace the path from the last deleted node towards the root
  - For each node x, check if AVL property ($-1 \leq bf(x) \leq 1$)
    - If yes
      - Proceed to $parent(x)$
    - If no,
      - Do an appropriate *rotation* at node x
      - Details (in next slide)
  - Continue to trace the path until we reach the root

What Rotation (after BST Deletion)?

- If AVL property is violated at node x
  - $If\ h(x.left) = h(x) - 1\ (that\ is, h(x.right) = h(x) - 3)$
    - $If\ h(x.left.left) = h(x) - 2 \rightarrow LL\ rotation(x)$
    - $Else\ If\ h(x.left.right) = h(x) - 2 \rightarrow LR\ rotation(x)$

  - $If\ h(x.right) = h(x) - 1\ (that\ is, h(x.left) = h(x) - 3)$
    - $If\ h(x.right.right) = h(x) - 2 \rightarrow RR\ rotation(x)$
    - $Else\ If\ h(x.right.left) = h(x) - 2 \rightarrow RL\ rotation(x)$

Note:

- The above is the same as in the insertion case

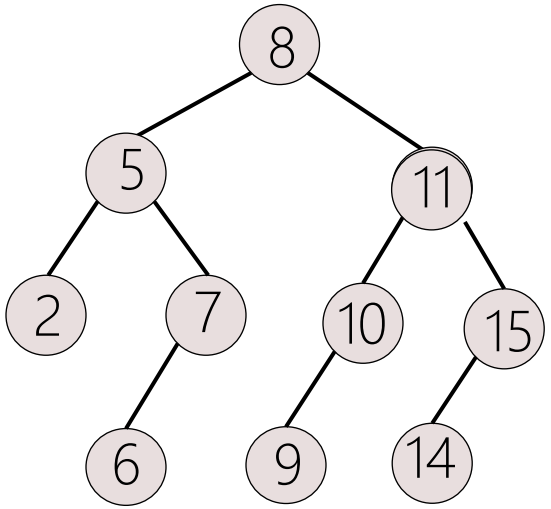
What Rotation?: Interpretation

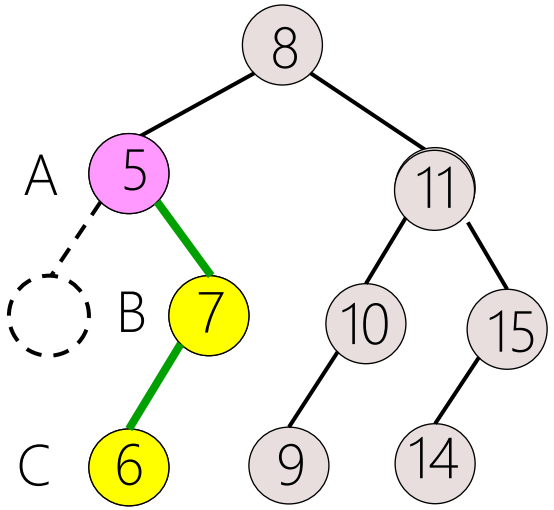
- For a given pivot (A),
  - Let B be the child of A with the longer height
  - Let C be the child of B with the longer height
    - If two children of B are of the same height, then C must be chosen in favor of single rotation

- Determine the type of imbalance based on $A - B - C$
    - *LL, LR, RR, or RL*

Delete (from Leaf): Example

- Delete 2 (leaf node)
  - Violation at node 5

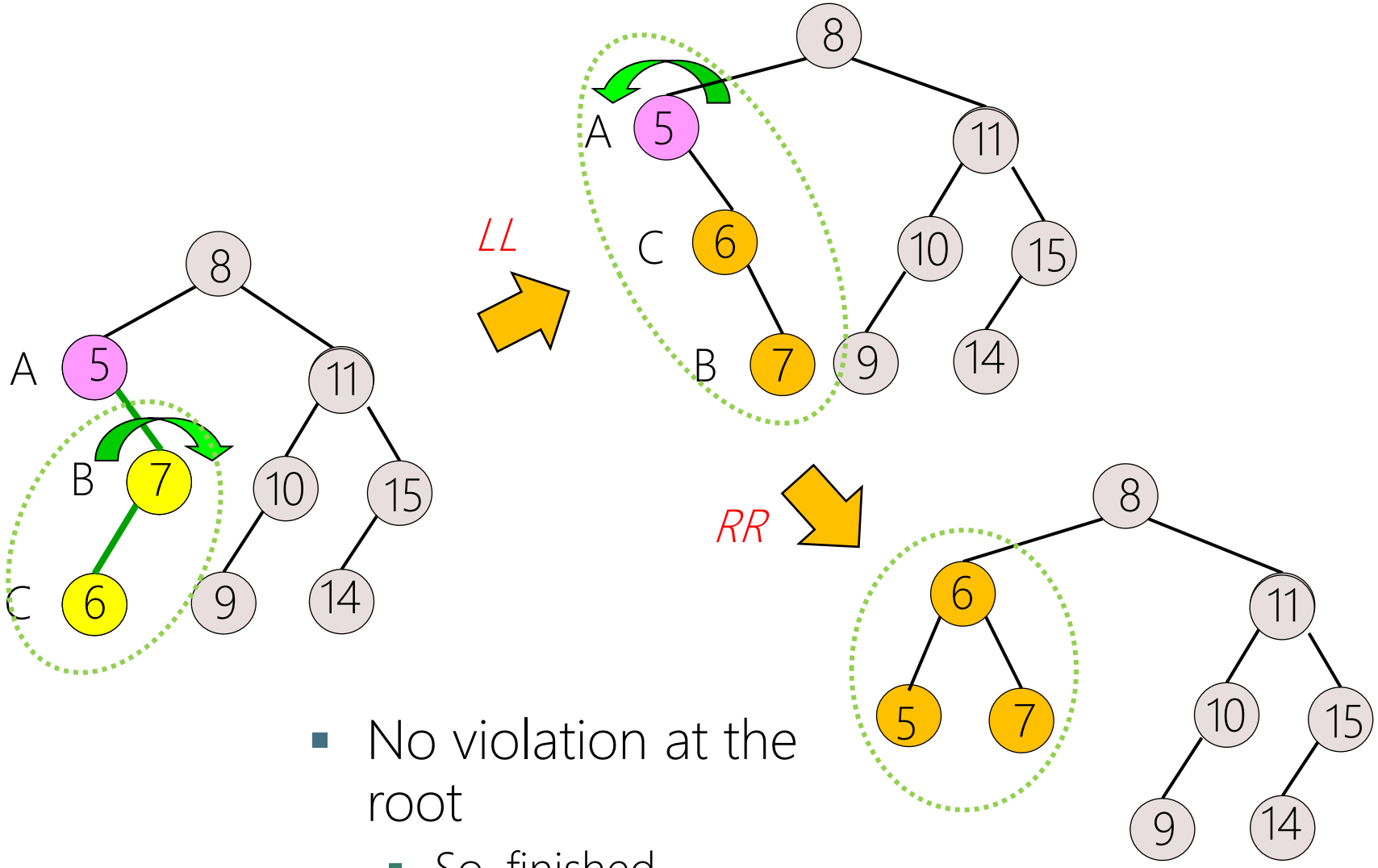




RL type of imbalance

Delete (from Leaf): Example

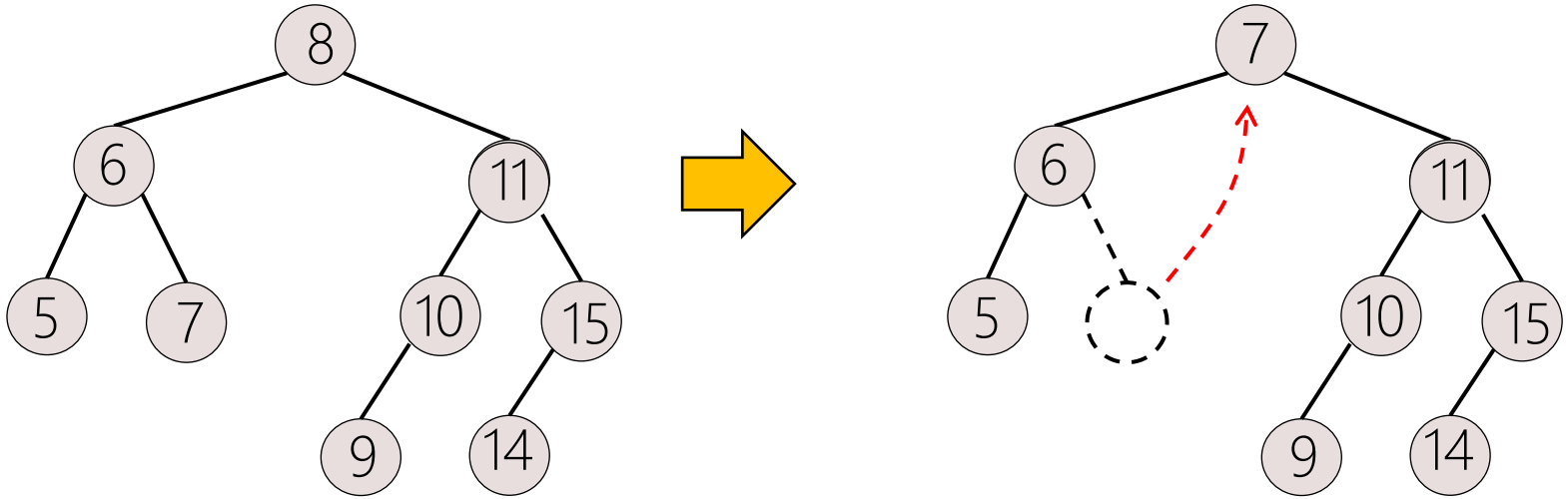
- RL rotation (= LL + RR)



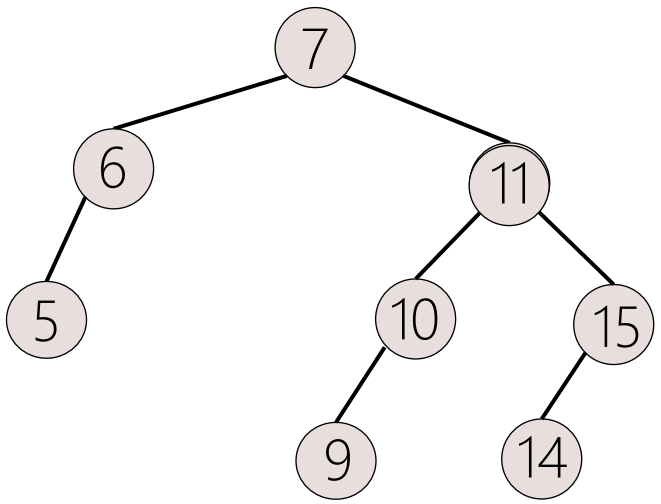
- No violation at the root
  - So, finished

Delete (from Deg-2 Node): Example

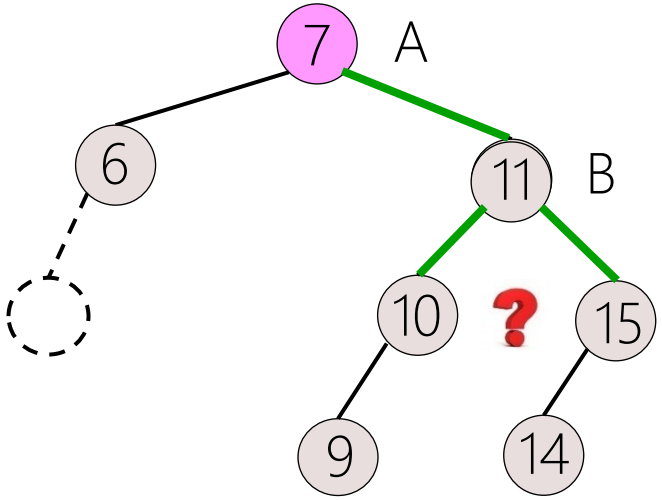
- Delete 8 (degree-2 node)
  - No violation at node 6, 7
  - So, finished



Delete: Children of the Same Height



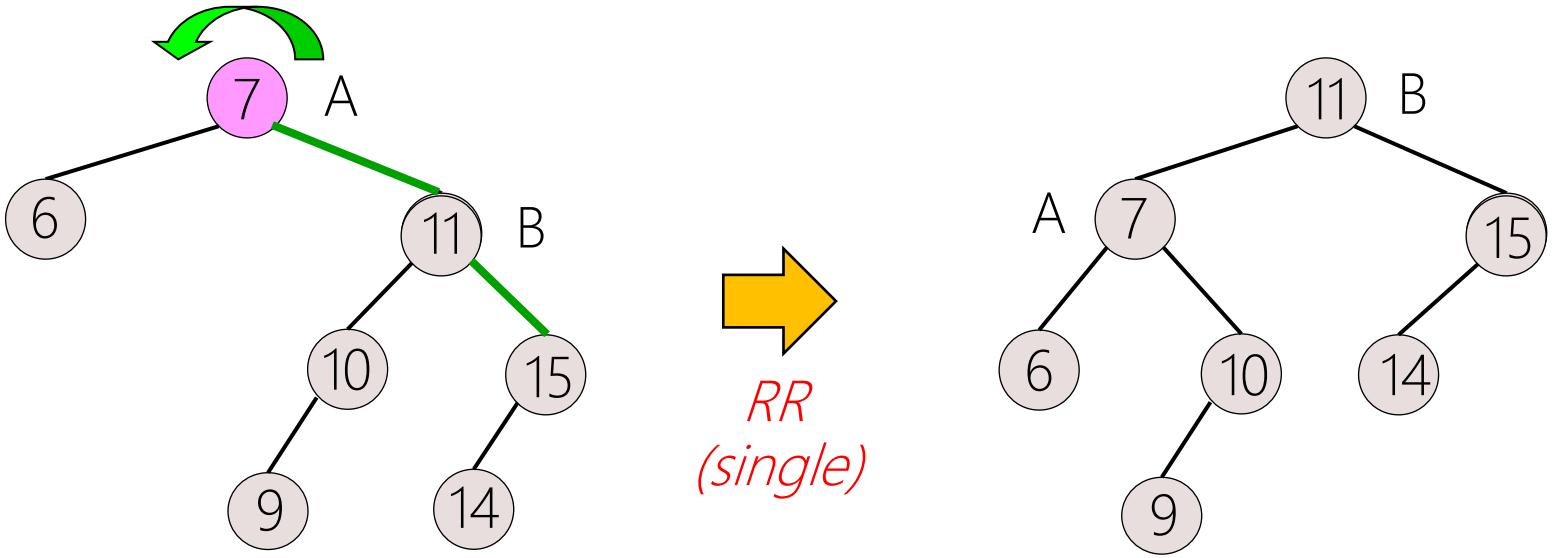
- Delete 5 (leaf node)
  - Violation at node 7



- Which type, RR or RL?(when two subtrees of B are of the same height)

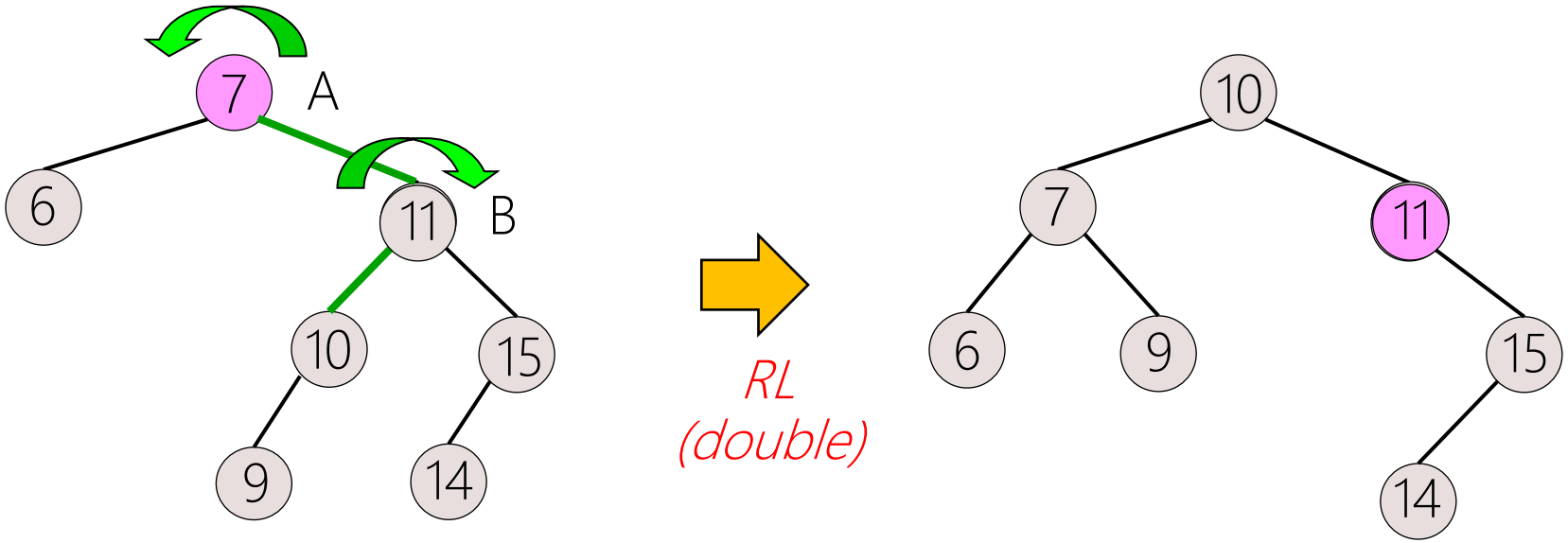
Delete: Children of the Same Height (1)

- When RR type (single rotation) is chosen:



Delete: Children of the Same Height (2)

- When RL type (double rotation) is chosen:



- Still violation at node 11

➔ Fail

Delete: Children of the Same Height (3)

- As a consequence
  - If two sub-trees of *B* are of the same height
  - Then a single rotation must be chosen

Quiz

- How many rotations can be required during one AVL deletion?

Quiz

- Give a scenario where a plain BST may outperform an AVL tree in practice, despite worse worst-case time complexity.

Summary

- Height-Balanced Tree
- Balance Factor
- Pivot Node
- Rotation
- LL / RR Imbalance
- LR / RL Imbalance
- Insertion Rebalancing
- Deletion Rebalancing